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# P8 · Chatter: Detecting social interaction in a room with non-visual modalities

Project 8 • Ziang Xue, Joseph Seok • November 10, 2026

**Question: can three phone microphones locate a speaker in a room?**

## Why it matters

Our goal is detecting conversation groups from audio only, on-device, with no camera. Knowing where each speaker stands is the first step.

## Setting and constraint

One indoor room. One person walking and reading aloud. Three separate single-mic phones on the floor. Constraint: localization accuracy.

## Success criterion

Median error ≤ 50 cm on a held-out recording, clearly below the no-audio baseline (98 cm).

*Status: everything shown is offline replay of three recorded sessions. No live runs or simulation yet.*

**Project page:** ziangxue.github.io/chatter **Slides PDF:** ziangxue.github.io/chatter/milestones/P08_Checkpoint.pdf

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# Approach: phone spectra → small model → position

## AUDIO PATH: WHAT THE SYSTEM USES

**3 phones**
2 iPhones, 1 iPad; synced by claps

→

**1 s windows**
16 kHz, 300–4000 Hz

→

**Spectral features**
8 bands or log-mel, relative to phone mean

→

**Small MLP / CNN**
6–12k parameters

→

**Position**
(x, y), or one 50 cm zone

## GROUND-TRUTH LABELS ONLY: CAMERA NEVER USED AT INFERENCE

**Wall camera**
1080p, 20 fps

→

**YOLO11m-pose**
Reused: ankle pixels

→

**Our calibration**
1.6 cm error on measured distances

→

**Floor position**
Label for every window

**Works and tested**
Tracker, sync, dataset (3 sessions, 2,963 windows), baselines, models

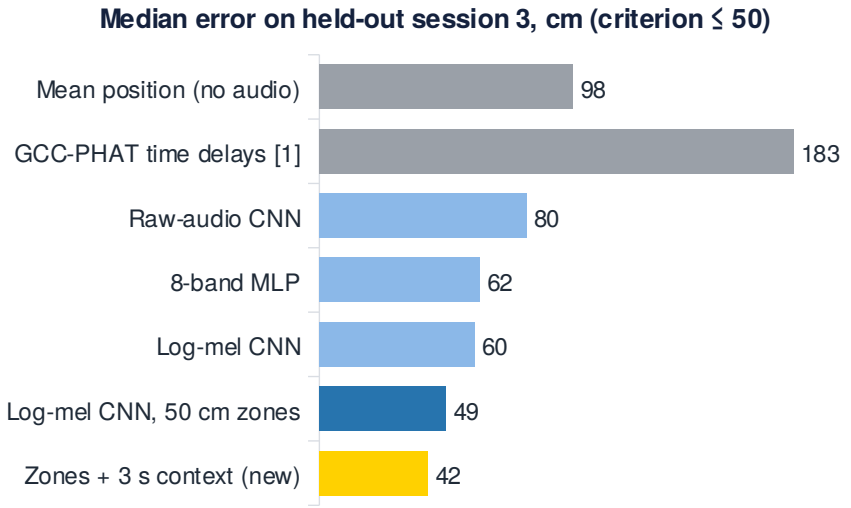
**Mode: recorded-data replay**
All results offline on recordings; no live inference or simulation yet

**Planned**
Several speakers, speaker counting, on-device deployment, live runs

**Key design choice:** compare per-band levels between phones, not arrival times, because room echo broke the timing (GCC-PHAT, slide 3).
**Built by us:** calibration, tracker pipeline, sync, dataset, all models and baselines. **Reused:** YOLO11m-pose (Ultralytics), PyTorch, SciPy.

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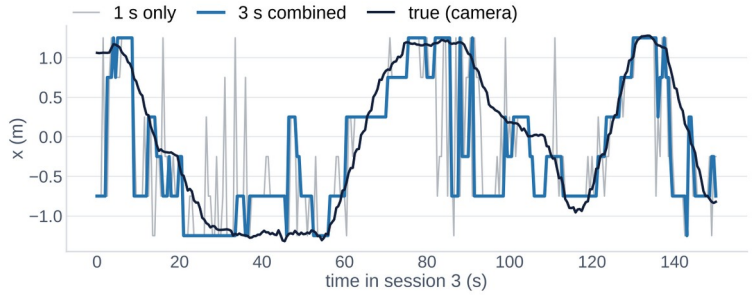
# Evidence: 49 cm, and 42 cm with 3 s of context




**NEW SINCE NOV 2: COMBINE 3 s OF ANSWERS**

**49 → 42 cm** gain 95% CI 4–11 cm

Past-only (live-usable): 46 cm, smaller gain




*Log-mel CNN, first 150 s of session 3: jumps removed*

**Supports:** position cues live in the frequency balance between phones; timing fails.

**Not shown:** other rooms; unwalked floor (82–90 cm); several speakers.

**Main uncertainty:** combining assumes one continuous speaker, as recorded; turn-taking and overlap are untested.

Preliminary, measured on recorded-data replay: train sessions 1–2, test once on session 3 (826 windows). CI: block bootstrap, 30 s blocks. Code: github.com/ZiangXue/chatter/tree/9f69ac3/feasibility/experiment

[1] Knapp & Carter, “The generalized correlation method for estimation of time delay,” IEEE Trans. ASSP, 1976.

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# Plan: localize and count speakers, then groups

| Evaluation | Plan |
|---|---|
| **Baselines** | Mean position / majority zone; GCC-PHAT |
| **Primary metric** | Median localization error (cm), held-out session |
| **Test data** | New recordings covering unwalked floor area |
| **Stress / ablation** | Two overlapping speakers; drop one phone |
| **Missing** | Labelled multi-speaker data; consent to record others |

**Minimum deliverable**
Audio-only localization and counting of 1–2 speakers

**Largest risk**
Overlapping speech breaks the spectral cue

**Fallback**
Add UWB tags as a second modality

**NEXT TASKS (OWNER, TARGET DATE)**

**Record unwalked floor area, claps at both ends; check tracker while walking**
Ziang · by Nov 14

**Done: 3 s smoothing 49 → 42 cm; dropping any phone +18–30 cm (all three needed)**
Joseph · done Nov 8

**Two-speaker test: simulate (pyroomacoustics), then record**
Joseph (simulation), Ziang (recording) · by Nov 19

**Our question for you:** Is ~50 cm localization enough to pursue conversation-group detection, or should we add UWB now?